Computer Architecture Lab 4

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1 Introduction

Memory access is very important if a lot of threads want to access memory. This can be optimized in multiple ways. It is important to consider in which order the memory is stored and what kind of memory should be used (global, shared or constant memory).

2 Coalesced memory access

Coalesced memory access is a technique which allows optimal usage of the global memory bandwidth. Since the used memory is close together, the GPU can optimize all of these memory requests. In regard to this exercise, where only the red component should be inverted of the image, the array can be rearranged as RR..RRGG..GGBB..BB. As a consequence this packs all the needed data closer together, optimizing the memory access. Fig 1 shows that using coalesced memory access is much faster than using uncoalesced memory (stored as RGB...RGB) access. In both cases the time scales linearly to the grid size.

3 Global, shared and constant memory

In this section, we explore the performance implications of utilizing different memory types (global, shared, and constant memory)for matrix multiplication. We design and implement kernels that multiply two matrices, assessing the processing time for each approach. The first approach operates solely on global memory, where each thread accesses matrix elements directly from device memory. This approach is straightforward but may lead to slower performance due to frequent memory accesses.
The second method is shared memory. Here, each thread loads a tile of matrices A and B into shared memory, enabling faster access during computation. By utilizing shared memory, which has lower latency and higher bandwidth compared to global memory, this method aims to reduce memory access times and improve performance.
Since matrices A and B are read-only and are likely to be accessed multiple times during the kernel's execution, storing them in constant memory can result in faster memory access times and enhance overall performance.

Figure 1: This image shows the timings of coalesced (orange) and uncoalesced memory access (blue) over 1000 data points per grid size

